CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 14: Equation of a Line

Topic-wise short notes with solved examples

Every straight line has an equation that all its points satisfy. This chapter covers the slope of a line, the conditions for parallel and perpendicular lines, and the different ways of writing the equation.

1 A point on a line

- The equation of a line is a relation between x and y that is satisfied by the coordinates of **every** point on the line, and by no other point.
- A point lies on a line if and only if its coordinates satisfy the equation of the line.

Example 1.1 Does the point $(2, -1)$ lie on the line $3x + 4y = 2$? Find k if $(k, 3)$ lies on this line.

SOLUTION

$3(2) + 4(-1) = 6 - 4 = 2$. The equation is satisfied, so the point lies on the line.

$3k + 4(3) = 2$ gives $3k = -10$

Yes; and $k = -\frac{10}{3}$.

2 Inclination and slope

- The **inclination** θ of a line is the angle it makes with the positive direction of the x-axis, measured anticlockwise.
- The **slope** (gradient) of a line is $m = \tan\theta$.

$$m = \tan\theta \qquad m = \frac{y_2 - y_1}{x_2 - x_1} \text{ for the line through } (x_1, y_1) \text{ and } (x_2, y_2)$$

Line	Inclination	Slope
Parallel to the x-axis	0°	0
Parallel to the y-axis	90°	Not defined
Inclined at 45°	45°	1
Inclined at 60°	60°	$\sqrt{3}$

Example 2.1 Find the slope and the inclination of the line through $A(2, 3)$ and $B(5, 6)$.

SOLUTION

$$m = \frac{6-3}{5-2} = 1$$

$$\tan\theta = 1$$

Slope $= 1$; inclination $= 45^\circ$

3 Parallel and perpendicular lines

- **Parallel** lines have equal slopes: $m_1 = m_2$.
- **Perpendicular** lines have slopes whose product is -1: $m_1 \times m_2 = -1$, so $m_2 = -\frac{1}{m_1}$.
- Three points A, B, C are **collinear** if slope of AB = slope of BC.

Example 3.1 Show that the points $A(1, 2)$, $B(3, 6)$ and $C(6, 12)$ are collinear.

SOLUTION

Slope of $AB = \frac{6-2}{3-1} = 2$ and slope of $BC = \frac{12-6}{6-3} = 2$

The slopes are equal and B is common, so the points are collinear.

Example 3.2 The line through $(2, 5)$ and $(4, k)$ is perpendicular to the line through $(1, 1)$ and $(5, 3)$. Find k.

SOLUTION

Slope of the second line $= \frac{3-1}{5-1} = \frac{1}{2}$, so the slope of the first must be -2.

$$\frac{k-5}{4-2} = -2, \text{ so } k - 5 = -4$$

$$k = 1$$

4 Forms of the equation

Form	Equation	Use it when you know
Slope-intercept	$y = mx + c$	The slope and the y-intercept c
Point-slope	$y - y_1 = m(x - x_1)$	The slope and one point
Two-point	$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$	Two points

- The **y-intercept** c is the y-coordinate of the point where the line cuts the y-axis. The **x-intercept** is the x-coordinate where it cuts the x-axis.
- $x = a$ is a line parallel to the y-axis and $y = b$ is a line parallel to the x-axis.

Example 4.1 Find the equation of the line with slope 2 that passes through $(1, -3)$.

SOLUTION

$y - (-3) = 2(x - 1)$

$y + 3 = 2x - 2$

$y = 2x - 5$

Example 4.2 Find the equation of the line through $(1, 2)$ and $(3, 8)$.

SOLUTION

Slope $= \dfrac{8 - 2}{3 - 1} = 3$

$y - 2 = 3(x - 1)$

$y = 3x - 1$

5 Slope and intercepts from an equation

- Write the equation in the form $y = mx + c$. Then m is the slope and c is the y-intercept.
- For $ax + by + c = 0$: slope $= -\dfrac{a}{b}$ and y-intercept $= -\dfrac{c}{b}$.
- For the x-intercept put $y = 0$; for the y-intercept put $x = 0$.

Example 5.1 Find the slope, the y-intercept and the x-intercept of the line $3x + 4y = 12$.

SOLUTION

$4y = -3x + 12$, so $y = -\dfrac{3}{4}x + 3$

Put $y = 0$: $3x = 12$, so $x = 4$

Slope $= -\dfrac{3}{4}$; y-intercept $= 3$; x-intercept $= 4$

6 Lines parallel or perpendicular to a given line

Example 6.1 Find the equation of the line through $(2, 1)$ that is (i) parallel (ii) perpendicular to the line $2x - 3y = 5$.

SOLUTION

The given line is $y = \dfrac{2}{3}x - \dfrac{5}{3}$, so its slope is $\dfrac{2}{3}$.

(i) Parallel line: slope $\dfrac{2}{3}$. $y - 1 = \dfrac{2}{3}(x - 2)$, so $3y - 3 = 2x - 4$

(ii) Perpendicular line: slope $-\dfrac{3}{2}$. $y - 1 = -\dfrac{3}{2}(x - 2)$, so $2y - 2 = -3x + 6$

(i) $2x - 3y - 1 = 0$ (ii) $3x + 2y - 8 = 0$

Example 6.2 Find the equation of the perpendicular bisector of the segment joining $A(2, 3)$ and $B(6, 5)$.

SOLUTION

Mid-point of $AB = (4, 4)$

Slope of $AB = \dfrac{5-3}{6-2} = \dfrac{1}{2}$, so the slope of the perpendicular bisector is -2.

$y - 4 = -2(x - 4)$

$2x + y = 12$

7 Common mistakes

- Finding the slope as $\dfrac{x_2 - x_1}{y_2 - y_1}$. It is the change in y over the change in x.
- Reading the slope from $3x + 4y = 12$ as 3. First make y the subject.
- Using $m_1 m_2 = 1$ for perpendicular lines. The product is -1.
- Forgetting that the perpendicular bisector passes through the **mid-point**.

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